

Lab: Designing a Basic Network Structure

Estimated time: **30** minutes

Objective: This lab introduces you to designing a basic network structure for a small organization using the drawing tool (Draw.io tool). The focus will be on understanding and placing essential network hardware components like routers, switches, and user devices.

Step 1: Introduction to basic network hardware

Task: Review the basic network components of a small organizational setup.

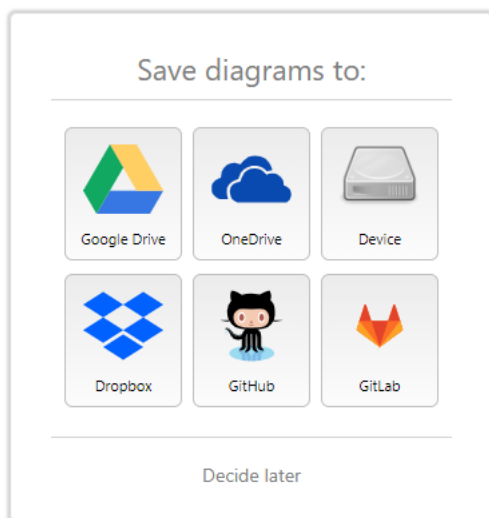
Components:

1. **Router:** Connects the organization's network to the internet.
2. **Switch:** Connects multiple devices within the local network.
3. **User Devices:** Employees use desktops, laptops, or printers.

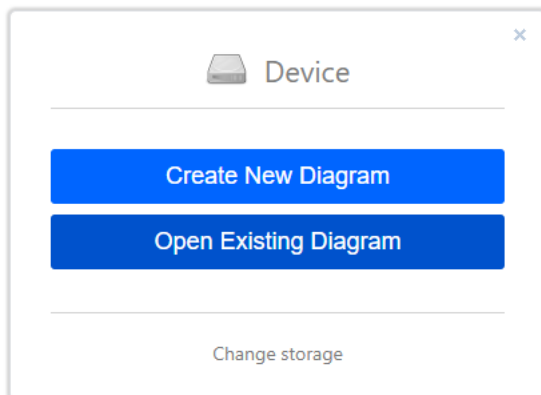
Step 2: Getting started with Draw.io:

Task 1: Open Draw.io and start a new blank diagram

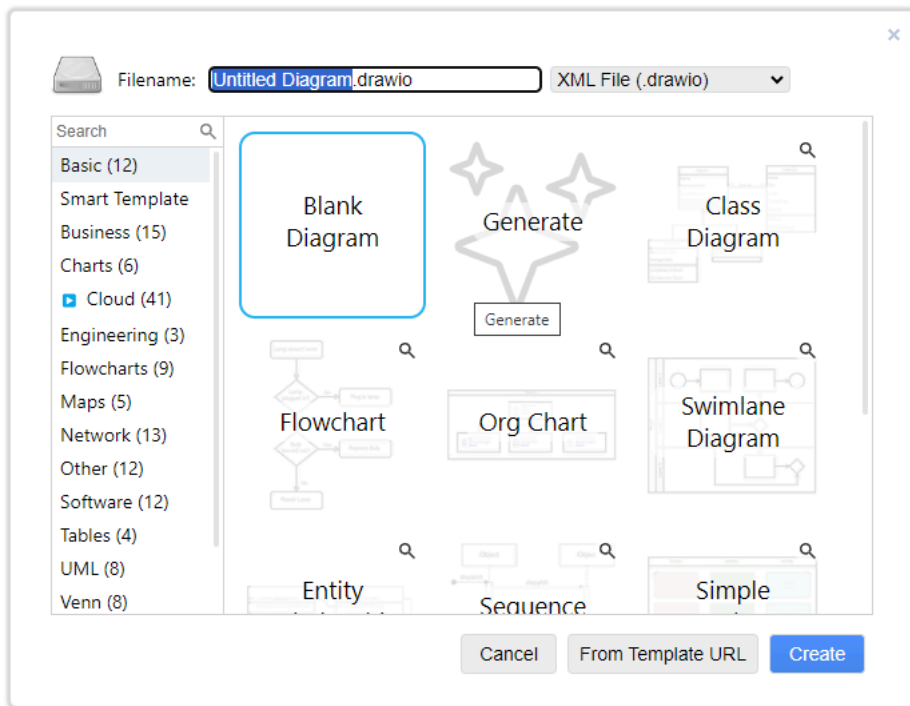
1. Open <https://app.diagrams.net/> and select your workspace to save the file.



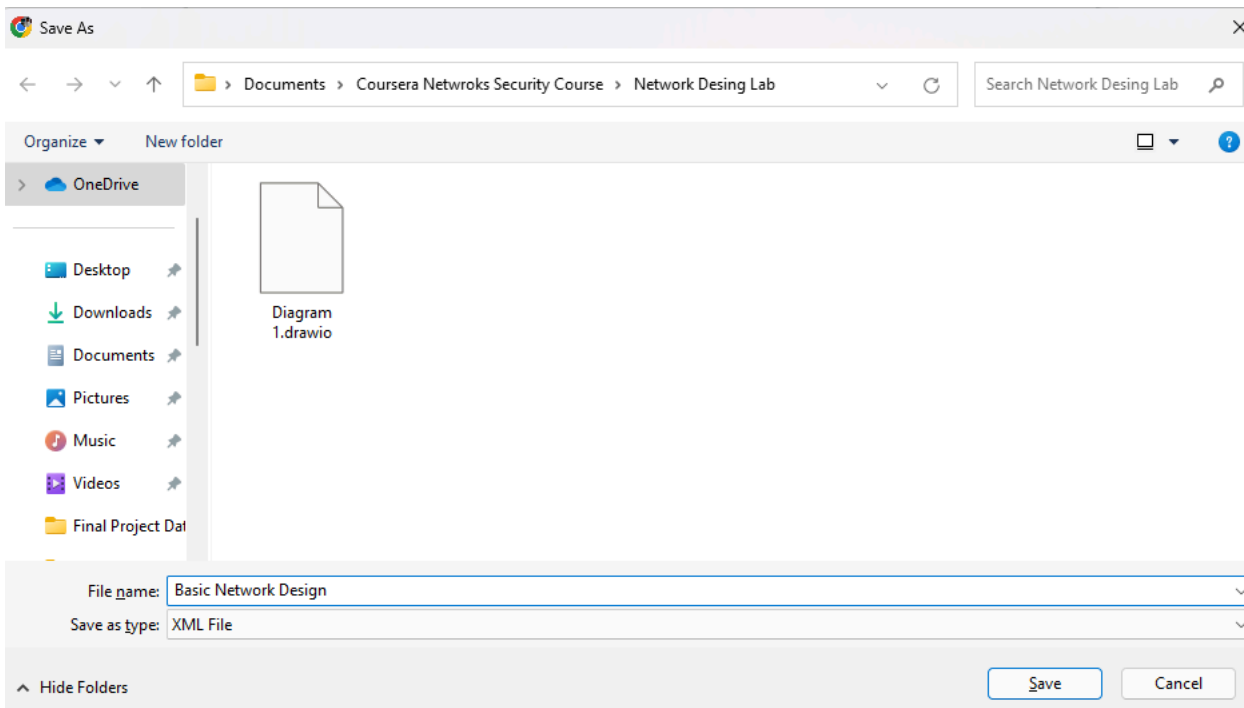
2. Select **Create New Diagram**.



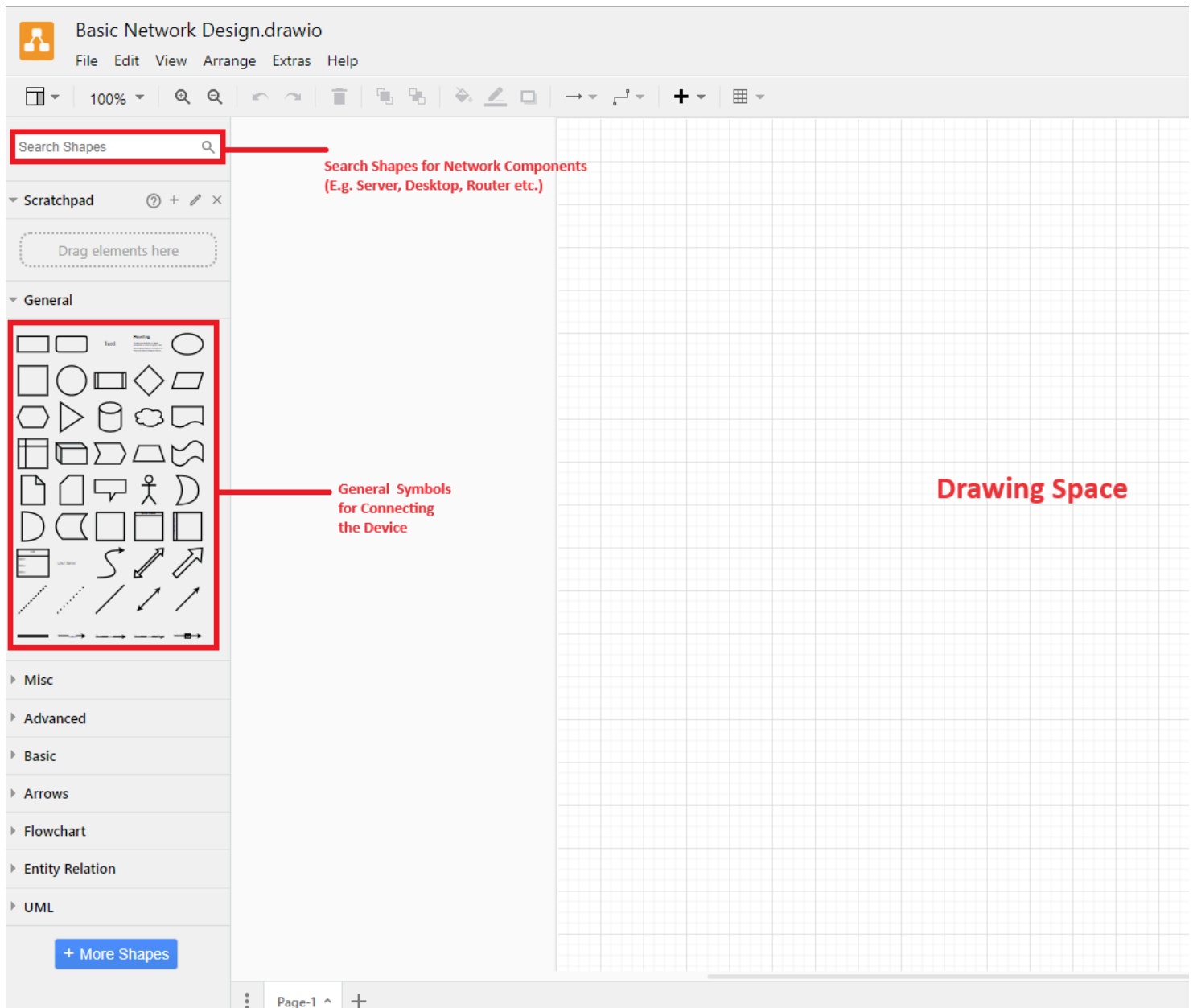
3. Choose a blank template and name your file (Basic Network Design) and click **Create**.



4. Save the file in your workspace (in this example, the selected workspace is the local system).



5. Familiarize yourself with the interface, focusing on the shape library where you'll find networking components.

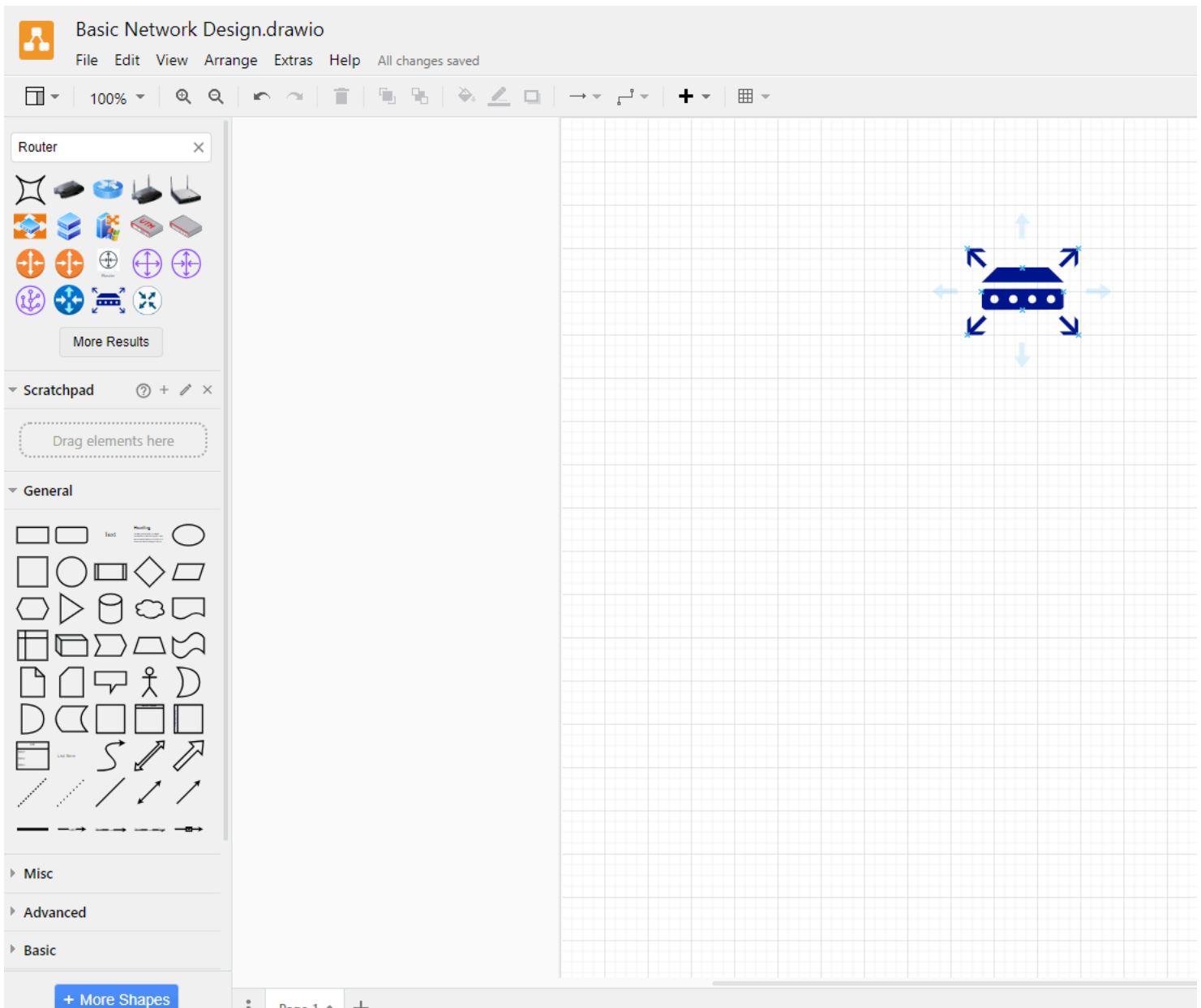


Step 3: Designing the basic network layout:

Task: Create a simple organizational network structure using basic network hardware

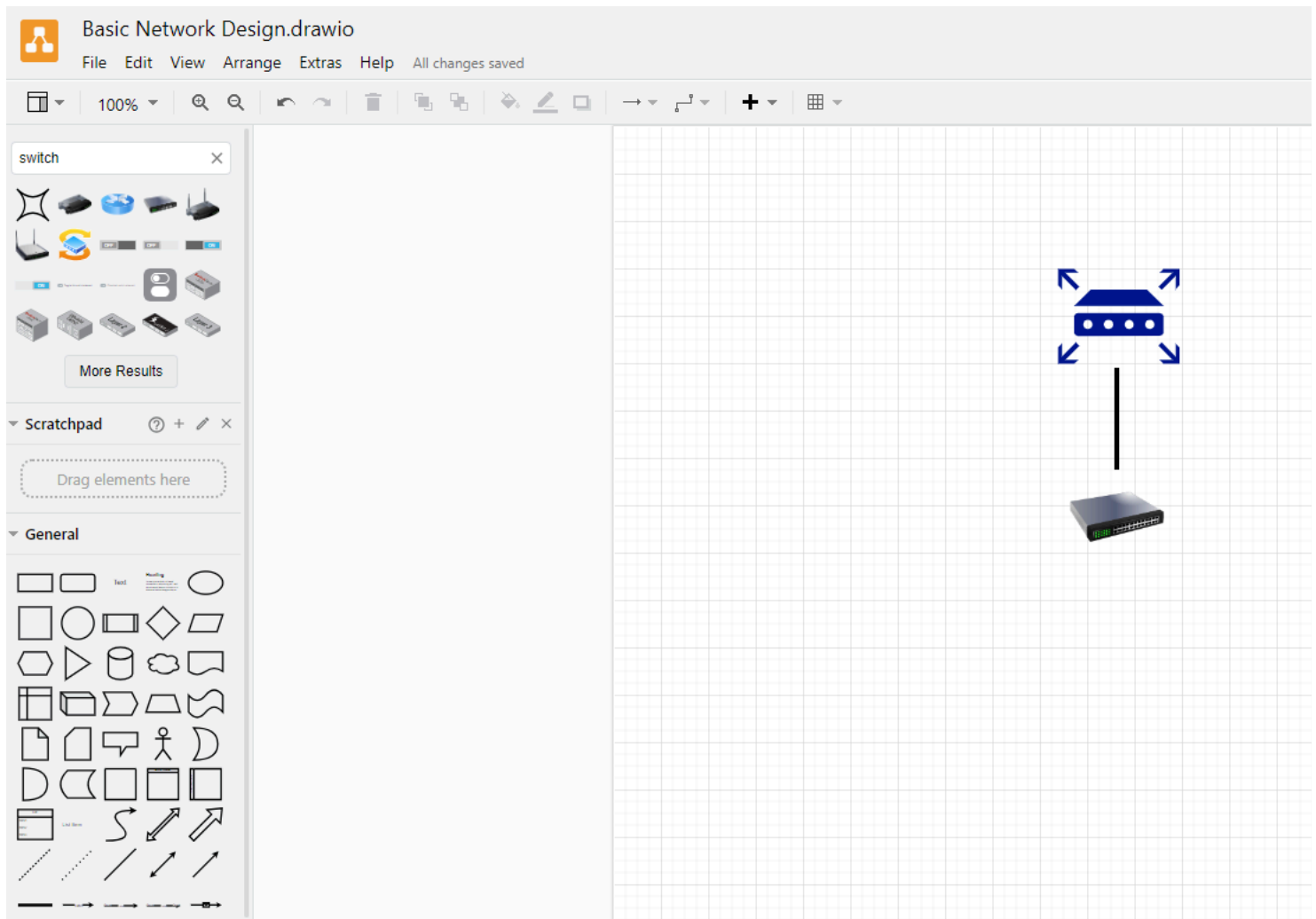
1. Adding the router

- Search for the router icon through the "Search Shape" option.
- Select the appropriate shape representing the router device and drag it to the drawing space.
- Place the router at the top, connecting the organization to the internet.



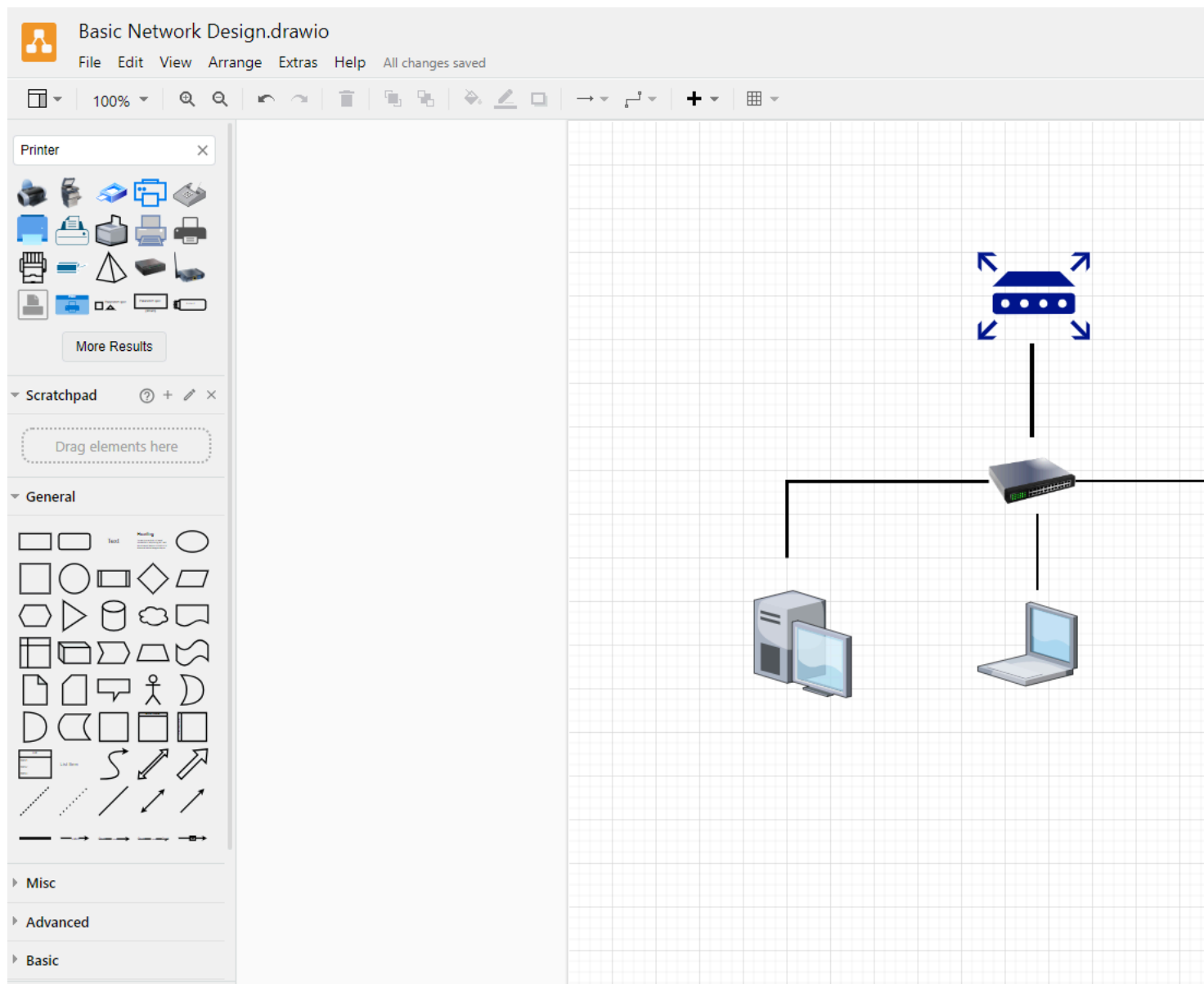
2. Adding the switch

- Search for the switch icon through the "Search Shape" option.
- Drag a switch icon following the router.
- Connect the router to the switch using the connector tool (line shapes in the general tab) to represent the link between them.



3. Adding user devices

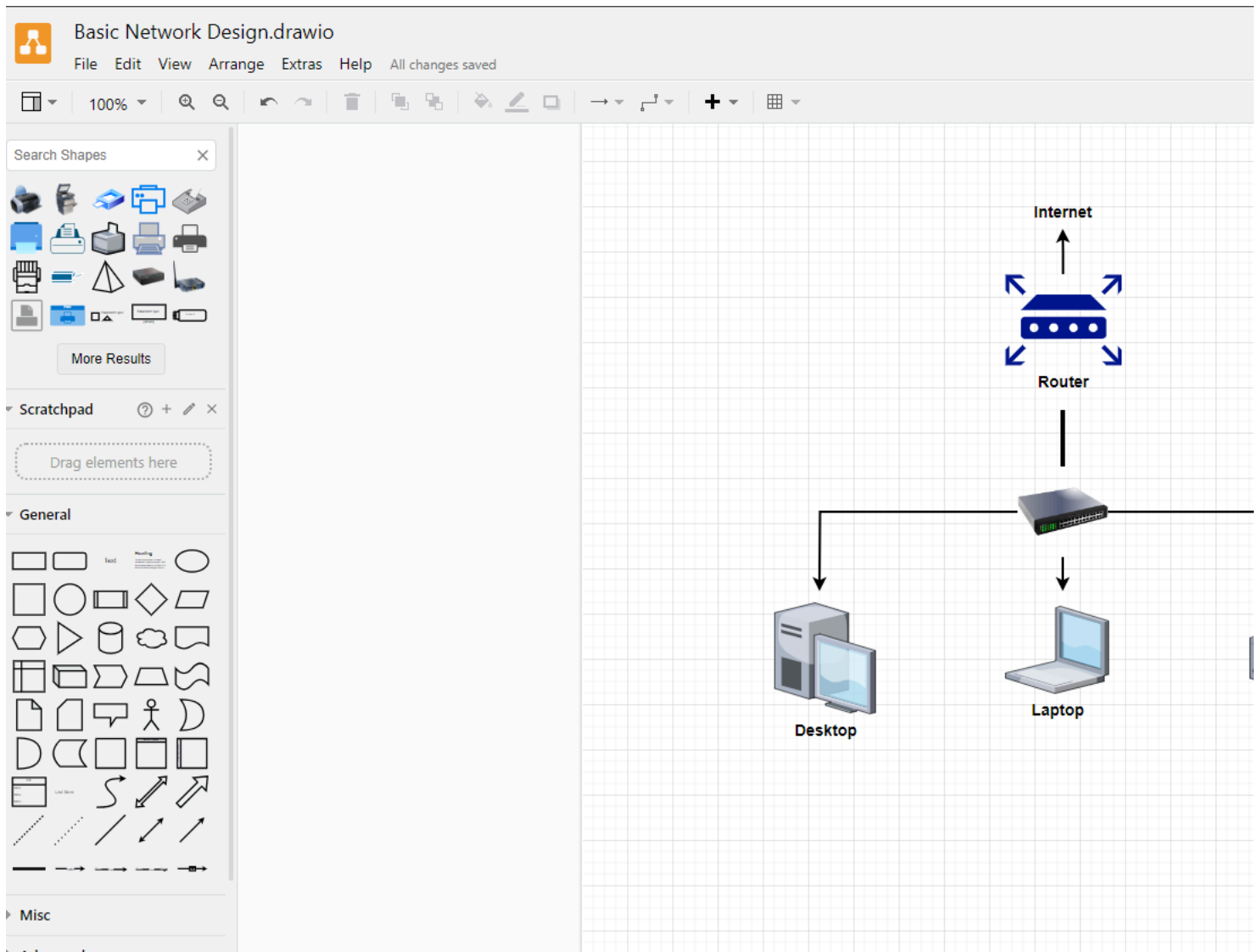
- Drag icons for desktops, laptops, or printers following the switch.
- Connect each device to the switch using the connector tool, showing how they share the network.



Step 4: Labeling and finalizing the diagram

Task: Ensure that all components are correctly labeled and connected.

- Double-click the component icons and add a label for each component ("Router," "Switch," "Employee PCs"). The text size and style can be adjusted with the options available in the right pane.
- Add a text box to represent the router's internet connection.
- Review the layout to ensure all devices are properly connected.



Step 5: Saving and exporting the diagram

- Click **File > Save As** to save your diagram to your local machine or cloud storage (The same options selected at the beginning of the project).
- Export the diagram as a PNG or PDF file for easy sharing or printing (**File > Export As > PNG/PDF**).

Exercises

Exercise 1: Designing a simple wired network

Objective:

Use essential network hardware to create a basic wired network layout for a small office.

Instructions:

1. Open <https://app.diagrams.net/> and start a new diagram.
2. Add the following components:
 - 1 Router
 - 1 Switch
 - 3 Desktops
3. Connect the router to the switch using a straight line.
4. Connect each desktop to the switch using straight lines.
5. Label each component as follows:
 - Router: "Office Router"
 - Switch: "Main Switch"
 - Desktops: "Employee PC 1," "Employee PC 2," and "Employee PC 3"
6. Save your diagram and export it as a PNG file.

Exercise 2: Adding wireless connectivity

Objective

Enhance the existing network design to include wireless connectivity for mobile devices.

Instructions

1. Open the diagram you created in Exercise 1.
 2. Add the following components:
 - 1 Wireless access point (WAP)
 - 2 Laptops
 3. Using a straight line, connect the wireless access point (WAP) to the switch.
 4. Indicate that the laptops are connected wirelessly to the WAP by drawing dotted lines between them.
 5. Label the new components as follows:
 - WAP: "Office WAP"
 - Laptops: "Employee Laptop 1" and "Employee Laptop 2"
 6. Save your updated diagram and export it as a PNG file.
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Summary:

In these exercises, you successfully designed a basic network layout for a small office using <https://app.diagrams.net/>, starting with a simple wired setup and then expanding it to wireless connectivity. These tasks helped reinforce the foundational concepts of network design, such as connecting essential hardware components like routers, switches, and wireless access points. By completing this lab, you have gained practical experience in visualizing and organizing network structures, which is crucial for managing and scaling real-world network environments.

Author(s)

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Skills Network